Toyota Eye Wear

Kexxu Prototype – Camera specifications

The prototype has 2 cameras, we'll refer to them as:

- Front Camera - for front world-view
- Eye Camera - for eye-tracking

For its specifications, since they are USB interfaced, we can use Video4Linux2 (V4L2) and the **USB utilities**.

Here are the results of specific CLIs for each camera

<u>Eye Camera</u>

- ## Interface ID and Name

  Lsusb

  ID 0c45:6366 Microdia Webcam Vitade AF

- ## Identify the Device Node

  v4l2-ctl --list-devices

  USB 2.0 Camera: USB Camera (usb-0000:00:14.0-9):

  /dev/video2

  /dev/video3

  /dev/media1

- ## Extract Video Specs (Resolutions & Frame Rates)

  v4l2-ctl -d /dev/video2 --list-formats-ext

  ioctl: VIDIOC_ENUM_FMT

  Type: Video Capture

  [0]: 'MJPG' (Motion-JPEG, compressed)

  Size: Discrete 1280x800

  Interval: Discrete 0.017s (60.000 fps)

  Interval: Discrete 0.033s (30.000 fps)

  Interval: Discrete 0.050s (20.000 fps)

  Interval: Discrete 0.067s (15.000 fps)

  Interval: Discrete 0.100s (10.000 fps)

  Interval: Discrete 0.200s (5.000 fps)

  Size: Discrete 1280x720

  Interval: Discrete 0.017s (60.000 fps)

  Interval: Discrete 0.033s (30.000 fps)

  Size: Discrete 640x480

  Interval: Discrete 0.017s (60.000 fps)

  Interval: Discrete 0.033s (30.000 fps)

  [1]: 'YUYV' (YUYV 4:2:2)

  Size: Discrete 640x480

  Interval: Discrete 0.033s (30.000 fps)

  Interval: Discrete 0.050s (20.000 fps)

  Interval: Discrete 0.067s (15.000 fps)

  Interval: Discrete 0.100s (10.000 fps)

Interval: Discrete 0.200s (5.000 fps)

Size: Discrete 1280x800

Interval: Discrete 0.100s (10.000 fps)

Interval: Discrete 0.200s (5.000 fps)

Size: Discrete 1280x720

Interval: Discrete 0.100s (10.000 fps)

Interval: Discrete 0.200s (5.000 fps)

- **Extract "Everything" via V4L2**

  v4l2-ctl -d /dev/video2 --all

  ```
  Driver Info:
  Driver name      : uvcvideo
  Card type        : USB 2.0 Camera: USB Camera
  Bus info         : usb-0000:00:14.0-9
  Driver version   : 6.8.12
  Capabilities     : 0x84a00001
  Video Capture
  Metadata Capture
  Streaming
  Extended Pix Format
  Device Capabilities
  Device Caps      : 0x04200001
  Video Capture
  Streaming
  Extended Pix Format
  Media Driver Info:
  Driver name      : uvcvideo
  Model            : USB 2.0 Camera: USB Camera
  Serial           : SN0001
  Bus info         : usb-0000:00:14.0-9
  Media version    : 6.8.12
  Hardware revision: 0x00000100 (256)
  ```

```
Driver version   : 6.8.12
Interface Info:
ID              : 0x03000002
Type            : V4L Video
Entity Info:
ID              : 0x00000001 (1)
Name            : USB 2.0 Camera: USB Camera
Function        : V4L2 I/O
Flags           : default
Pad 0x01000007   : 0: Sink
  Link 0x02000013: from remote pad 0x100000a of entity 'Extension 4' (Video
Pixel Formatter): Data, Enabled, Immutable
Priority: 2
Video input : 0 (Camera 1: ok)
Format Video Capture:
Width/Height      : 1280/800
Pixel Format      : 'MJPG' (Motion-JPEG)
Field             : None
Bytes per Line    : 0
Size Image        : 2048589
Colorspace        : sRGB
Transfer Function : Default (maps to sRGB)
YCbCr/HSV Encoding: Default (maps to ITU-R 601)
Quantization      : Default (maps to Full Range)
Flags             :
Crop Capability Video Capture:
Bounds      : Left 0, Top 0, Width 1280, Height 800
Default     : Left 0, Top 0, Width 1280, Height 800
Pixel Aspect: 1/1
Selection Video Capture: crop_default, Left 0, Top 0, Width 1280, Height 800,
Flags:
Selection Video Capture: crop_bounds, Left 0, Top 0, Width 1280, Height
800, Flags:
Streaming Parameters Video Capture:
Capabilities     : timeperframe
Frames per second: 25.000 (25/1)
Read buffers     : 0

User Controls
```

```
                    brightness 0x00980900 (int)    : min=-64 max=64 step=1
default=0 value=0
                     contrast 0x00980901 (int)    : min=0 max=64 step=1 default=32
value=32
                    saturation 0x00980902 (int)    : min=0 max=128 step=1
default=64 value=64
                       hue 0x00980903 (int)    : min=-40 max=40 step=1 default=0
value=0
       white_balance_automatic 0x0098090c (bool)   : default=1 value=1
                      gamma 0x00980910 (int)    : min=72 max=500 step=1
default=100 value=100
                       gain 0x00980913 (int)    : min=0 max=100 step=1 default=0
value=0
          power_line_frequency 0x00980918 (menu)   : min=0 max=2 default=1
value=1 (50 Hz)
0: Disabled
1: 50 Hz
2: 60 Hz
     white_balance_temperature 0x0098091a (int)    : min=2800 max=6500
step=1 default=4600 value=4600 flags=inactive
                    sharpness 0x0098091b (int)    : min=0 max=6 step=1 default=3
value=3
        backlight_compensation 0x0098091c (int)    : min=0 max=2 step=1
default=1 value=1

Camera Controls

                 auto_exposure 0x009a0901 (menu)   : min=0 max=3 default=3
value=3 (Aperture Priority Mode)
1: Manual Mode
3: Aperture Priority Mode
        exposure_time_absolute 0x009a0902 (int)    : min=1 max=5000 step=1
default=157 value=157 flags=inactive
    exposure_dynamic_framerate 0x009a0903 (bool)   : default=0 value=1
```

- **Available hardware controls (Brightness, Contrast, Focus, Exposure ranges)**

```
v4l2-ctl -d /dev/video2 --list-ctrls
User Controls

                     brightness 0x00980900 (int)    : min=-64 max=64 step=1 default=0
value=0
                       contrast 0x00980901 (int)    : min=0 max=64 step=1 default=32
value=32
                     saturation 0x00980902 (int)    : min=0 max=128 step=1 default=64
value=64
                            hue 0x00980903 (int)    : min=-40 max=40 step=1 default=0
value=0
        white_balance_automatic 0x0098090c (bool)   : default=1 value=1
                          gamma 0x00980910 (int)    : min=72 max=500 step=1
default=100 value=100
                           gain 0x00980913 (int)    : min=0 max=100 step=1 default=0
value=0
           power_line_frequency 0x00980918 (menu)   : min=0 max=2 default=1
value=1 (50 Hz)
      white_balance_temperature 0x0098091a (int)    : min=2800 max=6500
step=1 default=4600 value=4600 flags=inactive
                      sharpness 0x0098091b (int)    : min=0 max=6 step=1 default=3
value=3
         backlight_compensation 0x0098091c (int)    : min=0 max=2 step=1
default=1 value=1

Camera Controls

                  auto_exposure 0x009a0901 (menu)   : min=0 max=3 default=3
value=3 (Aperture Priority Mode)
         exposure_time_absolute 0x009a0902 (int)    : min=1 max=5000 step=1
default=157 value=157 flags=inactive
```

```
exposure_dynamic_framerate 0x009a0903 (bool)   : default=0 value=1
```

- ## Extract Hardware & Descriptor Details

```
sudo lsusb -v -d 0c45:6366

Device Descriptor:
  bLength                18
  bDescriptorType         1
  bcdUSB               2.00
  bDeviceClass          239 Miscellaneous Device
  bDeviceSubClass         2 [unknown]
  bDeviceProtocol         1 Interface Association
  bMaxPacketSize0        64
  idVendor           0x0c45 Microdia
  idProduct          0x6366 Webcam Vitade AF
  bcdDevice            1.00
  iManufacturer           2 Sonix Technology Co., Ltd.
  iProduct                1 USB 2.0 Camera
  iSerial                 3 SN0001
  bNumConfigurations      1
  Configuration Descriptor:
    bLength                 9
    bDescriptorType         2
    wTotalLength       0x021f
    bNumInterfaces          2
    bConfigurationValue     1
    iConfiguration          0
    bmAttributes         0x80
      (Bus Powered)
    MaxPower              500mA
    Interface Association:
      bLength                 8
      bDescriptorType        11
      bFirstInterface         0
      bInterfaceCount         2
      bFunctionClass         14 Video
      bFunctionSubClass       3 Video Interface Collection
      bFunctionProtocol       0
```

```
  iFunction               5 USB Camera
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
  bInterfaceNumber        0
  bAlternateSetting       0
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      1 Video Control
  bInterfaceProtocol      0
  iInterface              5 USB Camera
  VideoControl Interface Descriptor:
    bLength               13
    bDescriptorType       36
    bDescriptorSubtype     1 (HEADER)
    bcdUVC                1.00
    wTotalLength          0x006d
    dwClockFrequency      15.000000MHz
    bInCollection          1
    baInterfaceNr( 0)      1
  VideoControl Interface Descriptor:
    bLength                9
    bDescriptorType       36
    bDescriptorSubtype     3 (OUTPUT_TERMINAL)
    bTerminalID            5
    wTerminalType      0x0101 USB Streaming
    bAssocTerminal         0
    bSourceID              4
    iTerminal              0
  VideoControl Interface Descriptor:
    bLength               29
    bDescriptorType       36
    bDescriptorSubtype     6 (EXTENSION_UNIT)
    bUnitID                3
    guidExtensionCode        {28f03370-6311-4a2e-ba2c-6890eb334016}
    bNumControls          32
    bNrInPins              1
    baSourceID( 0)         2
    bControlSize           4
    bmControls( 0)      0xff
```

```
    bmControls( 1)       0xff
    bmControls( 2)       0xff
    bmControls( 3)       0xff
    iExtension           0
  VideoControl Interface Descriptor:
    bLength              29
    bDescriptorType      36
    bDescriptorSubtype   6 (EXTENSION_UNIT)
    bUnitID              4
    guidExtensionCode    {00000000-0000-0000-0000-000000000000}
    bNumControls         32
    bNrInPins            1
    baSourceID( 0)       3
    bControlSize         4
    bmControls( 0)       0x07
    bmControls( 1)       0x80
    bmControls( 2)       0x3f
    bmControls( 3)       0x9f
    iExtension           0
  VideoControl Interface Descriptor:
    bLength              18
    bDescriptorType      36
    bDescriptorSubtype   2 (INPUT_TERMINAL)
    bTerminalID          1
    wTerminalType        0x0201 Camera Sensor
    bAssocTerminal       0
    iTerminal            0
    wObjectiveFocalLengthMin      0
    wObjectiveFocalLengthMax      0
    wOcularFocalLength            0
    bControlSize                  3
    bmControls           0x0000000e
      Auto-Exposure Mode
      Auto-Exposure Priority
      Exposure Time (Absolute)
  VideoControl Interface Descriptor:
    bLength              11
    bDescriptorType      36
    bDescriptorSubtype   5 (PROCESSING_UNIT)
  Warning: Descriptor too short
```

```
    bUnitID                2
    bSourceID              1
    wMaxMultiplier         0
    bControlSize           2
    bmControls     0x0000177f
      Brightness
      Contrast
      Hue
      Saturation
      Sharpness
      Gamma
      White Balance Temperature
      Backlight Compensation
      Gain
      Power Line Frequency
      White Balance Temperature, Auto
    iProcessing            0
    bmVideoStandards     0x00
  Endpoint Descriptor:
    bLength                7
    bDescriptorType        5
    bEndpointAddress     0x83  EP 3 IN
    bmAttributes           3
      Transfer Type            Interrupt
      Synch Type               None
      Usage Type               Data
    wMaxPacketSize     0x0010  1x 16 bytes
    bInterval              6
Interface Descriptor:
  bLength                9
  bDescriptorType        4
  bInterfaceNumber       1
  bAlternateSetting      0
  bNumEndpoints          0
  bInterfaceClass       14 Video
  bInterfaceSubClass     2 Video Streaming
  bInterfaceProtocol     0
  iInterface             0
  VideoStreaming Interface Descriptor:
    bLength                         15
```

```
    bDescriptorType         36
    bDescriptorSubtype        1 (INPUT_HEADER)
    bNumFormats             2
    wTotalLength        0x0123
    bEndpointAddress        0x81  EP 1 IN
    bmInfo                  0
    bTerminalLink           5
    bStillCaptureMethod       0
    bTriggerSupport         0
    bTriggerUsage           0
    bControlSize            1
    bmaControls( 0)         0
    bmaControls( 1)         0
  VideoStreaming Interface Descriptor:
    bLength                 11
    bDescriptorType         36
    bDescriptorSubtype        6 (FORMAT_MJPEG)
    bFormatIndex            1
    bNumFrameDescriptors        3
    bFlags                  0
      Fixed-size samples: No
    bDefaultFrameIndex        1
    bAspectRatioX           0
    bAspectRatioY           0
    bmInterlaceFlags        0x00
      Interlaced stream or variable: No
      Fields per frame: 1 fields
      Field 1 first: No
      Field pattern: Field 1 only
    bCopyProtect            0
  VideoStreaming Interface Descriptor:
    bLength                 50
    bDescriptorType         36
    bDescriptorSubtype        7 (FRAME_MJPEG)
    bFrameIndex             1
    bmCapabilities          0x00
      Still image unsupported
    wWidth                  1280
    wHeight                 800
    dwMinBitRate            81920000
```

```
  dwMaxBitRate                983040000
  dwMaxVideoFrameBufferSize     2048589
  dwDefaultFrameInterval         333333
  bFrameIntervalType                  6
  dwFrameInterval( 0)            166666
  dwFrameInterval( 1)            333333
  dwFrameInterval( 2)            500000
  dwFrameInterval( 3)            666666
  dwFrameInterval( 4)           1000000
  dwFrameInterval( 5)           2000000
VideoStreaming Interface Descriptor:
  bLength                            34
  bDescriptorType                    36
  bDescriptorSubtype                  7 (FRAME_MJPEG)
  bFrameIndex                         2
  bmCapabilities                   0x00
    Still image unsupported
  wWidth                           1280
  wHeight                           720
  dwMinBitRate                442368000
  dwMaxBitRate                884736000
  dwMaxVideoFrameBufferSize     1843789
  dwDefaultFrameInterval         333333
  bFrameIntervalType                  2
  dwFrameInterval( 0)            166666
  dwFrameInterval( 1)            333333
VideoStreaming Interface Descriptor:
  bLength                            34
  bDescriptorType                    36
  bDescriptorSubtype                  7 (FRAME_MJPEG)
  bFrameIndex                         3
  bmCapabilities                   0x00
    Still image unsupported
  wWidth                            640
  wHeight                           480
  dwMinBitRate                147456000
  dwMaxBitRate                294912000
  dwMaxVideoFrameBufferSize      614989
  dwDefaultFrameInterval         333333
  bFrameIntervalType                  2
```

```
      dwFrameInterval( 0)          166666
      dwFrameInterval( 1)          333333
    VideoStreaming Interface Descriptor:
      bLength                         27
      bDescriptorType                 36
      bDescriptorSubtype               4 (FORMAT_UNCOMPRESSED)
      bFormatIndex                     2
      bNumFrameDescriptors             3
      guidFormat                       {32595559-0000-0010-8000-
00aa00389b71}
      bBitsPerPixel                   16
      bDefaultFrameIndex               1
      bAspectRatioX                    0
      bAspectRatioY                    0
      bmInterlaceFlags                0x00
        Interlaced stream or variable: No
        Fields per frame: 2 fields
        Field 1 first: No
        Field pattern: Field 1 only
      bCopyProtect                     0
    VideoStreaming Interface Descriptor:
      bLength                         46
      bDescriptorType                 36
      bDescriptorSubtype               5 (FRAME_UNCOMPRESSED)
      bFrameIndex                      1
      bmCapabilities                  0x00
        Still image unsupported
      wWidth                         640
      wHeight                        480
      dwMinBitRate                24576000
      dwMaxBitRate                147456000
      dwMaxVideoFrameBufferSize      614400
      dwDefaultFrameInterval         333333
      bFrameIntervalType               5
      dwFrameInterval( 0)          333333
      dwFrameInterval( 1)          500000
      dwFrameInterval( 2)          666666
      dwFrameInterval( 3)         1000000
      dwFrameInterval( 4)         2000000
    VideoStreaming Interface Descriptor:
```

```
    bLength                             34
    bDescriptorType                     36
    bDescriptorSubtype                   5 (FRAME_UNCOMPRESSED)
    bFrameIndex                          2
    bmCapabilities                    0x00
      Still image unsupported
    wWidth                            1280
    wHeight                            800
    dwMinBitRate                    81920000
    dwMaxBitRate                   163840000
    dwMaxVideoFrameBufferSize        2048000
    dwDefaultFrameInterval           1000000
    bFrameIntervalType                   2
    dwFrameInterval( 0)              1000000
    dwFrameInterval( 1)              2000000
  VideoStreaming Interface Descriptor:
    bLength                             34
    bDescriptorType                     36
    bDescriptorSubtype                   5 (FRAME_UNCOMPRESSED)
    bFrameIndex                          3
    bmCapabilities                    0x00
      Still image unsupported
    wWidth                            1280
    wHeight                            720
    dwMinBitRate                    73728000
    dwMaxBitRate                   147456000
    dwMaxVideoFrameBufferSize        1843200
    dwDefaultFrameInterval           1000000
    bFrameIntervalType                   2
    dwFrameInterval( 0)              1000000
    dwFrameInterval( 1)              2000000
  VideoStreaming Interface Descriptor:
    bLength                              6
    bDescriptorType                     36
    bDescriptorSubtype                  13 (COLORFORMAT)
    bColorPrimaries                      1 (BT.709,sRGB)
    bTransferCharacteristics             1 (BT.709)
    bMatrixCoefficients                  4 (SMPTE 170M (BT.601))
Interface Descriptor:
  bLength                 9
```

```
    bDescriptorType         4
    bInterfaceNumber        1
    bAlternateSetting       1
    bNumEndpoints           1
    bInterfaceClass        14 Video
    bInterfaceSubClass      2 Video Streaming
    bInterfaceProtocol      0
    iInterface              0
    Endpoint Descriptor:
      bLength                 7
      bDescriptorType         5
      bEndpointAddress     0x81  EP 1 IN
      bmAttributes            5
        Transfer Type            Isochronous
        Synch Type               Asynchronous
        Usage Type               Data
      wMaxPacketSize     0x0080  1x 128 bytes
      bInterval               1
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        1
    bAlternateSetting       2
    bNumEndpoints           1
    bInterfaceClass        14 Video
    bInterfaceSubClass      2 Video Streaming
    bInterfaceProtocol      0
    iInterface              0
    Endpoint Descriptor:
      bLength                 7
      bDescriptorType         5
      bEndpointAddress     0x81  EP 1 IN
      bmAttributes            5
        Transfer Type            Isochronous
        Synch Type               Asynchronous
        Usage Type               Data
      wMaxPacketSize     0x0100  1x 256 bytes
      bInterval               1
  Interface Descriptor:
    bLength                 9
```

```
  bDescriptorType         4
  bInterfaceNumber        1
  bAlternateSetting       3
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
  iInterface              0
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes            5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0320  1x 800 bytes
    bInterval               1
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
  bInterfaceNumber        1
  bAlternateSetting       4
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
  iInterface              0
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes            5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0b20  2x 800 bytes
    bInterval               1
Interface Descriptor:
  bLength                 9
```

```
      bDescriptorType         4
      bInterfaceNumber        1
      bAlternateSetting       5
      bNumEndpoints           1
      bInterfaceClass        14 Video
      bInterfaceSubClass      2 Video Streaming
      bInterfaceProtocol      0
      iInterface              0
      Endpoint Descriptor:
        bLength                 7
        bDescriptorType         5
        bEndpointAddress     0x81  EP 1 IN
        bmAttributes            5
          Transfer Type            Isochronous
          Synch Type               Asynchronous
          Usage Type               Data
        wMaxPacketSize     0x1320  3x 800 bytes
        bInterval               1
    Interface Descriptor:
      bLength                 9
      bDescriptorType         4
      bInterfaceNumber        1
      bAlternateSetting       6
      bNumEndpoints           1
      bInterfaceClass        14 Video
      bInterfaceSubClass      2 Video Streaming
      bInterfaceProtocol      0
      iInterface              0
      Endpoint Descriptor:
        bLength                 7
        bDescriptorType         5
        bEndpointAddress     0x81  EP 1 IN
        bmAttributes            5
          Transfer Type            Isochronous
          Synch Type               Asynchronous
          Usage Type               Data
        wMaxPacketSize     0x1400  3x 1024 bytes
        bInterval               1
  Device Qualifier (for other device speed):
    bLength                10
```

```
    bDescriptorType         6
    bcdUSB               2.00
    bDeviceClass          239 Miscellaneous Device
    bDeviceSubClass         2 [unknown]
    bDeviceProtocol         1 Interface Association
    bMaxPacketSize0        64
    bNumConfigurations      1
  Device Status:     0x0002
    (Bus Powered)
    Remote Wakeup Enabled
```

- ## Query System & Kernel Attributes

  (specific serial number or UUID for a udev rule or script)
  udevadm info --query=all --name=/dev/video2

```
P: /devices/pci0000:00/0000:00:14.0/usb1/1-9/1-9:1.0/video4linux/video2
M: video2
R: 2
U: video4linux
D: c 81:2
N: video2
L: 0
S: v4l/by-id/usb-Sonix_Technology_Co.__Ltd._USB_2.0_Camera_SN0001-video-index0
S: v4l/by-path/pci-0000:00:14.0-usb-0:9:1.0-video-index0
S: v4l/by-path/pci-0000:00:14.0-usbv2-0:9:1.0-video-index0
E: DEVPATH=/devices/pci0000:00/0000:00:14.0/usb1/1-9/1-9:1.0/video4linux/video2
E: DEVNAME=/dev/video2
E: MAJOR=81
E: MINOR=2
E: SUBSYSTEM=video4linux
E: USEC_INITIALIZED=1830222113
E: ID_V4L_VERSION=2
E: ID_V4L_PRODUCT=USB 2.0 Camera: USB Camera
E: ID_V4L_CAPABILITIES=:capture:
E: ID_BUS=usb
E: ID_MODEL=USB_2.0_Camera
E: ID_MODEL_ENC=USB\x202.0\x20Camera
```

E: ID_MODEL_ID=6366
E: ID_SERIAL=Sonix_Technology_Co.__Ltd._USB_2.0_Camera_SN0001
E: ID_SERIAL_SHORT=SN0001
E: ID_VENDOR=Sonix_Technology_Co.__Ltd.
E: ID_VENDOR_ENC=Sonix\x20Technology\x20Co.\x2c\x20Ltd.
E: ID_VENDOR_ID=0c45
E: ID_REVISION=0100
E: ID_TYPE=video
E: ID_USB_MODEL=USB_2.0_Camera
E: ID_USB_MODEL_ENC=USB\x202.0\x20Camera
E: ID_USB_MODEL_ID=6366
E: ID_USB_SERIAL=Sonix_Technology_Co.__Ltd._USB_2.0_Camera_SN0001
E: ID_USB_SERIAL_SHORT=SN0001
E: ID_USB_VENDOR=Sonix_Technology_Co.__Ltd.
E: ID_USB_VENDOR_ENC=Sonix\x20Technology\x20Co.\x2c\x20Ltd.
E: ID_USB_VENDOR_ID=0c45
E: ID_USB_REVISION=0100
E: ID_USB_TYPE=video
E: ID_USB_INTERFACES=:0e0100:0e0200:
E: ID_USB_INTERFACE_NUM=00
E: ID_USB_DRIVER=uvcvideo
E: ID_PATH_WITH_USB_REVISION=pci-0000:00:14.0-usbv2-0:9:1.0
E: ID_PATH=pci-0000:00:14.0-usb-0:9:1.0
E: ID_PATH_TAG=pci-0000_00_14_0-usb-0_9_1_0
E: ID_FOR_SEAT=video4linux-pci-0000_00_14_0-usb-0_9_1_0
E: COLORD_DEVICE=1
E: COLORD_KIND=camera
E: DEVLINKS=/dev/v4l/by-id/usb-Sonix_Technology_Co.__Ltd._USB_2.0_Camera_SN0001-video-index0 /dev/v4l/by-path/pci-0000:00:14.0-usb-0:9:1.0-video-index0 /dev/v4l/by-path/pci-0000:00:14>
E: TAGS=:seat:uaccess:snap_firefox_firefox:snap_firefox_geckodriver:
E: CURRENT_TAGS=:seat:uaccess:snap_firefox_firefox:snap_firefox_geckodriver:

Front Camera

- ## Interface ID and Name

lsusb

ID 0bda:5842 Realtek Semiconductor Corp. ICT Camera

- ## Identify the Device Node

  v4l2-ctl --list-devices

ICT Camera: ICT Camera (usb-0000:00:14.0-9):

/dev/video2

/dev/video3

/dev/media1

- ## Extract Video Specs (Resolutions & Frame Rates)

  v4l2-ctl -d /dev/video2 --list-formats-ext

ioctl: VIDIOC_ENUM_FMT

Type: Video Capture

[0]: 'MJPG' (Motion-JPEG, compressed)

Size: Discrete 1920x1080

Interval: Discrete 0.033s (30.000 fps)

Size: Discrete 1280x720

Interval: Discrete 0.033s (30.000 fps)

[1]: 'YUYV' (YUYV 4:2:2)

Size: Discrete 1920x1080

Interval: Discrete 0.200s (5.000 fps)

Size: Discrete 1280x720

Interval: Discrete 0.100s (10.000 fps)

- **Extract "Everything" via V4L2**

  v4l2-ctl -d /dev/video2 --all

  ```
  Driver Info:
  Driver name      : uvcvideo
  Card type        : ICT Camera: ICT Camera
  Bus info         : usb-0000:00:14.0-9
  Driver version   : 6.8.12
  Capabilities     : 0x84a00001
  Video Capture
  Metadata Capture
  Streaming
  Extended Pix Format
  Device Capabilities
  Device Caps      : 0x04200001
  Video Capture
  Streaming
  Extended Pix Format
  Media Driver Info:
  Driver name      : uvcvideo
  Model            : ICT Camera: ICT Camera
  Serial           : 200901010001
  Bus info         : usb-0000:00:14.0-9
  Media version    : 6.8.12
  Hardware revision: 0x00000001 (1)
  Driver version   : 6.8.12
  Interface Info:
  ID               : 0x03000002
  Type             : V4L Video
  Entity Info:
  ID               : 0x00000001 (1)
  Name             : ICT Camera: ICT Camera
  Function         : V4L2 I/O
  Flags            : default
  ```

```
Pad 0x01000007   : 0: Sink
  Link 0x02000010: from remote pad 0x100000a of entity 'Extension 4' (Video
Pixel Formatter): Data, Enabled, Immutable
Priority: 2
Video input : 0 (Camera 1: ok)
Format Video Capture:
Width/Height      : 1920/1080
Pixel Format      : 'MJPG' (Motion-JPEG)
Field             : None
Bytes per Line    : 0
Size Image        : 4147200
Colorspace        : sRGB
Transfer Function : Rec. 709
YCbCr/HSV Encoding: ITU-R 601
Quantization      : Default (maps to Full Range)
Flags             :
Crop Capability Video Capture:
Bounds      : Left 0, Top 0, Width 1920, Height 1080
Default     : Left 0, Top 0, Width 1920, Height 1080
Pixel Aspect: 1/1
Selection Video Capture: crop_default, Left 0, Top 0, Width 1920, Height
1080, Flags:
Selection Video Capture: crop_bounds, Left 0, Top 0, Width 1920, Height
1080, Flags:
Streaming Parameters Video Capture:
Capabilities     : timeperframe
Frames per second: 30.000 (30/1)
Read buffers     : 0

User Controls

                    brightness 0x00980900 (int)    : min=-64 max=64 step=1
default=-8 value=-8
                      contrast 0x00980901 (int)    : min=0 max=100 step=1
default=60 value=60
                    saturation 0x00980902 (int)    : min=0 max=128 step=1
default=54 value=54
                         hue 0x00980903 (int)    : min=-180 max=180 step=1
default=0 value=0
       white_balance_automatic 0x0098090c (bool)   : default=1 value=1
```

gamma 0x00980910 (int) : min=100 max=500 step=1 default=300 value=300
power_line_frequency 0x00980918 (menu) : min=0 max=2 default=1 value=1 (50 Hz)
0: Disabled
1: 50 Hz
2: 60 Hz
white_balance_temperature 0x0098091a (int) : min=2800 max=6500 step=10 default=4600 value=4600 flags=inactive
sharpness 0x0098091b (int) : min=0 max=100 step=1 default=65 value=65
backlight_compensation 0x0098091c (int) : min=0 max=2 step=1 default=1 value=1

Camera Controls

auto_exposure 0x009a0901 (menu) : min=0 max=3 default=3 value=3 (Aperture Priority Mode)
1: Manual Mode
3: Aperture Priority Mode
exposure_time_absolute 0x009a0902 (int) : min=1 max=10000 step=1 default=166 value=166 flags=inactive
exposure_dynamic_framerate 0x009a0903 (bool) : default=0 value=0
pan_absolute 0x009a0908 (int) : min=-57600 max=57600 step=3600 default=0 value=0
tilt_absolute 0x009a0909 (int) : min=-43200 max=43200 step=3600 default=0 value=0
zoom_absolute 0x009a090d (int) : min=0 max=3 step=1 default=0 value=0

- **Available hardware controls (Brightness, Contrast, Focus, Exposure ranges)**

v4l2-ctl -d /dev/video2 --list-ctrls

User Controls

```
                brightness 0x00980900 (int)    : min=-64 max=64 step=1 default=-8
value=-8
                  contrast 0x00980901 (int)    : min=0 max=100 step=1 default=60
value=60
                saturation 0x00980902 (int)    : min=0 max=128 step=1 default=54
value=54
                     hue 0x00980903 (int)    : min=-180 max=180 step=1 default=0
value=0
      white_balance_automatic 0x0098090c (bool)   : default=1 value=1
                    gamma 0x00980910 (int)    : min=100 max=500 step=1
default=300 value=300
         power_line_frequency 0x00980918 (menu)   : min=0 max=2 default=1
value=1 (50 Hz)
     white_balance_temperature 0x0098091a (int)    : min=2800 max=6500
step=10 default=4600 value=4600 flags=inactive
                 sharpness 0x0098091b (int)    : min=0 max=100 step=1 default=65
value=65
        backlight_compensation 0x0098091c (int)    : min=0 max=2 step=1
default=1 value=1
```

Camera Controls

```
               auto_exposure 0x009a0901 (menu)   : min=0 max=3 default=3
value=3 (Aperture Priority Mode)
        exposure_time_absolute 0x009a0902 (int)    : min=1 max=10000 step=1
default=166 value=166 flags=inactive
    exposure_dynamic_framerate 0x009a0903 (bool)   : default=0 value=0
                pan_absolute 0x009a0908 (int)    : min=-57600 max=57600
step=3600 default=0 value=0
               tilt_absolute 0x009a0909 (int)    : min=-43200 max=43200 step=3600
default=0 value=0
               zoom_absolute 0x009a090d (int)    : min=0 max=3 step=1 default=0
value=0
```

- ## Extract Hardware & Descriptor Details

sudo lsusb -v -d 0bda:5842

```
Device Descriptor:
  bLength                18
  bDescriptorType         1
  bcdUSB               2.00
  bDeviceClass          239 Miscellaneous Device
  bDeviceSubClass         2 [unknown]
  bDeviceProtocol         1 Interface Association
  bMaxPacketSize0        64
  idVendor           0x0bda Realtek Semiconductor Corp.
  idProduct          0x5842 ICT Camera
  bcdDevice            0.01
  iManufacturer           3 Generic
  iProduct                1 ICT Camera
  iSerial                 2 200901010001
  bNumConfigurations      1
  Configuration Descriptor:
    bLength                 9
    bDescriptorType         2
    wTotalLength       0x01c2
    bNumInterfaces          2
    bConfigurationValue     1
    iConfiguration          4 ICT Camera
```

```
bmAttributes         0x80
  (Bus Powered)
MaxPower             500mA
Interface Association:
  bLength                8
  bDescriptorType        11
  bFirstInterface        0
  bInterfaceCount        2
  bFunctionClass         14 Video
  bFunctionSubClass      3 Video Interface Collection
  bFunctionProtocol      0
  iFunction              5 ICT Camera
Interface Descriptor:
  bLength                9
  bDescriptorType        4
  bInterfaceNumber       0
  bAlternateSetting      0
  bNumEndpoints          1
  bInterfaceClass        14 Video
  bInterfaceSubClass     1 Video Control
  bInterfaceProtocol     0
  iInterface             5 ICT Camera
  VideoControl Interface Descriptor:
    bLength              13
    bDescriptorType      36
    bDescriptorSubtype   1 (HEADER)
    bcdUVC               1.00
```

```
  wTotalLength       0x004e
  dwClockFrequency       15.000000MHz
  bInCollection          1
  baInterfaceNr( 0)      1
VideoControl Interface Descriptor:
  bLength                18
  bDescriptorType        36
  bDescriptorSubtype     2 (INPUT_TERMINAL)
  bTerminalID            1
  wTerminalType      0x0201 Camera Sensor
  bAssocTerminal         0
  iTerminal              0
  wObjectiveFocalLengthMin      0
  wObjectiveFocalLengthMax      0
  wOcularFocalLength            0
  bControlSize                  3
  bmControls          0x00002a0e
    Auto-Exposure Mode
    Auto-Exposure Priority
    Exposure Time (Absolute)
    Zoom (Absolute)
    PanTilt (Absolute)
    Roll (Absolute)
VideoControl Interface Descriptor:
  bLength                11
  bDescriptorType        36
  bDescriptorSubtype     5 (PROCESSING_UNIT)
```

```
Warning: Descriptor too short
  bUnitID                 2
  bSourceID               1
  wMaxMultiplier          0
  bControlSize            2
  bmControls     0x0000157f
    Brightness
    Contrast
    Hue
    Saturation
    Sharpness
    Gamma
    White Balance Temperature
    Backlight Compensation
    Power Line Frequency
    White Balance Temperature, Auto
  iProcessing             0
  bmVideoStandards     0x09
    None
    SECAM - 625/50
VideoControl Interface Descriptor:
  bLength                 9
  bDescriptorType        36
  bDescriptorSubtype      3 (OUTPUT_TERMINAL)
  bTerminalID             3
  wTerminalType      0x0101 USB Streaming
  bAssocTerminal          0
```

```
    bSourceID               4
    iTerminal               0
  VideoControl Interface Descriptor:
    bLength                 27
    bDescriptorType         36
    bDescriptorSubtype      6 (EXTENSION_UNIT)
    bUnitID                 4
    guidExtensionCode       {1229a78c-47b4-4094-b0ce-db07386fb938}
    bNumControls            2
    bNrInPins               1
    baSourceID( 0)          2
    bControlSize            2
    bmControls( 0)          0x00
    bmControls( 1)          0x06
    iExtension              0
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress        0x83  EP 3 IN
    bmAttributes            3
      Transfer Type         Interrupt
      Synch Type            None
      Usage Type            Data
    wMaxPacketSize          0x0010  1x 16 bytes
    bInterval               6
Interface Descriptor:
  bLength                   9
```

```
    bDescriptorType         4
    bInterfaceNumber        1
    bAlternateSetting       0
    bNumEndpoints           0
    bInterfaceClass        14 Video
    bInterfaceSubClass      2 Video Streaming
    bInterfaceProtocol      0
    iInterface              0
    VideoStreaming Interface Descriptor:
      bLength                 15
      bDescriptorType         36
      bDescriptorSubtype       1 (INPUT_HEADER)
      bNumFormats              2
      wTotalLength        0x00d5
      bEndpointAddress        0x81  EP 1 IN
      bmInfo                  0
      bTerminalLink           3
      bStillCaptureMethod      2
      bTriggerSupport          1
      bTriggerUsage            0
      bControlSize             1
      bmaControls( 0)          0
      bmaControls( 1)          0
    VideoStreaming Interface Descriptor:
      bLength                 11
      bDescriptorType         36
      bDescriptorSubtype       6 (FORMAT_MJPEG)
```

```
    bFormatIndex                    1
    bNumFrameDescriptors            2
    bFlags                          1
      Fixed-size samples: Yes
    bDefaultFrameIndex              1
    bAspectRatioX                   0
    bAspectRatioY                   0
    bmInterlaceFlags                0x00
      Interlaced stream or variable: No
      Fields per frame: 1 fields
      Field 1 first: No
      Field pattern: Field 1 only
    bCopyProtect                    0
  VideoStreaming Interface Descriptor:
    bLength                         30
    bDescriptorType                 36
    bDescriptorSubtype              7 (FRAME_MJPEG)
    bFrameIndex                     1
    bmCapabilities                  0x00
      Still image unsupported
    wWidth                          1920
    wHeight                         1080
    dwMinBitRate                    995328000
    dwMaxBitRate                    995328000
    dwMaxVideoFrameBufferSize       4147200
    dwDefaultFrameInterval          333333
    bFrameIntervalType              1
```

```
  dwFrameInterval( 0)           333333
VideoStreaming Interface Descriptor:
  bLength                        30
  bDescriptorType                36
  bDescriptorSubtype              7 (FRAME_MJPEG)
  bFrameIndex                     2
  bmCapabilities                 0x00
    Still image unsupported
  wWidth                       1280
  wHeight                       720
  dwMinBitRate                442368000
  dwMaxBitRate                442368000
  dwMaxVideoFrameBufferSize     1843200
  dwDefaultFrameInterval         333333
  bFrameIntervalType              1
  dwFrameInterval( 0)           333333
VideoStreaming Interface Descriptor:
  bLength                        14
  bDescriptorType                36
  bDescriptorSubtype              3 (STILL_IMAGE_FRAME)
  bEndpointAddress               0x00  EP 0 OUT
  bNumImageSizePatterns           2
  wWidth( 0)                    1920
  wHeight( 0)                   1080
  wWidth( 1)                    1280
  wHeight( 1)                    720
  bNumCompressionPatterns         0
```

```
VideoStreaming Interface Descriptor:
  bLength                 6
  bDescriptorType        36
  bDescriptorSubtype      13 (COLORFORMAT)
  bColorPrimaries         1 (BT.709,sRGB)
  bTransferCharacteristics  1 (BT.709)
  bMatrixCoefficients     4 (SMPTE 170M (BT.601))
VideoStreaming Interface Descriptor:
  bLength                27
  bDescriptorType        36
  bDescriptorSubtype      4 (FORMAT_UNCOMPRESSED)
  bFormatIndex           2
  bNumFrameDescriptors     2
  guidFormat             {32595559-0000-0010-8000-00aa00389b71}
  bBitsPerPixel          16
  bDefaultFrameIndex       1
  bAspectRatioX          0
  bAspectRatioY          0
  bmInterlaceFlags       0x00
    Interlaced stream or variable: No
    Fields per frame: 2 fields
    Field 1 first: No
    Field pattern: Field 1 only
  bCopyProtect           0
VideoStreaming Interface Descriptor:
  bLength                30
  bDescriptorType        36
```

```
    bDescriptorSubtype               5 (FRAME_UNCOMPRESSED)
    bFrameIndex                      1
    bmCapabilities                   0x00
      Still image unsupported
    wWidth                           1920
    wHeight                          1080
    dwMinBitRate                     165888000
    dwMaxBitRate                     165888000
    dwMaxVideoFrameBufferSize        4147200
    dwDefaultFrameInterval           2000000
    bFrameIntervalType               1
    dwFrameInterval( 0)              2000000
  VideoStreaming Interface Descriptor:
    bLength                          30
    bDescriptorType                  36
    bDescriptorSubtype               5 (FRAME_UNCOMPRESSED)
    bFrameIndex                      2
    bmCapabilities                   0x00
      Still image unsupported
    wWidth                           1280
    wHeight                          720
    dwMinBitRate                     147456000
    dwMaxBitRate                     147456000
    dwMaxVideoFrameBufferSize        1843200
    dwDefaultFrameInterval           1000000
    bFrameIntervalType               1
    dwFrameInterval( 0)              1000000
```

```
  VideoStreaming Interface Descriptor:
    bLength                             14
    bDescriptorType                     36
    bDescriptorSubtype                   3 (STILL_IMAGE_FRAME)
    bEndpointAddress                    0x00  EP 0 OUT
    bNumImageSizePatterns                2
    wWidth( 0)                        1920
    wHeight( 0)                       1080
    wWidth( 1)                        1280
    wHeight( 1)                        720
    bNumCompressionPatterns              0
  VideoStreaming Interface Descriptor:
    bLength                              6
    bDescriptorType                     36
    bDescriptorSubtype                  13 (COLORFORMAT)
    bColorPrimaries                     1 (BT.709,sRGB)
    bTransferCharacteristics             1 (BT.709)
    bMatrixCoefficients                  4 (SMPTE 170M (BT.601))
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
  bInterfaceNumber        1
  bAlternateSetting       1
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
```

```
  iInterface             0
  Endpoint Descriptor:
    bLength                7
    bDescriptorType        5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes           5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0080  1x 128 bytes
    bInterval              1
Interface Descriptor:
  bLength                9
  bDescriptorType        4
  bInterfaceNumber       1
  bAlternateSetting      2
  bNumEndpoints          1
  bInterfaceClass       14 Video
  bInterfaceSubClass     2 Video Streaming
  bInterfaceProtocol     0
  iInterface             0
  Endpoint Descriptor:
    bLength                7
    bDescriptorType        5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes           5
      Transfer Type            Isochronous
```

```
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0200  1x 512 bytes
    bInterval               1
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
  bInterfaceNumber        1
  bAlternateSetting       3
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
  iInterface              0
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes            5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0400  1x 1024 bytes
    bInterval               1
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
```

```
  bInterfaceNumber        1
  bAlternateSetting       4
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
  iInterface              0
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes            5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0b00  2x 768 bytes
    bInterval               1
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
  bInterfaceNumber        1
  bAlternateSetting       5
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
  iInterface              0
```

```
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes            5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
      Usage Type               Data
    wMaxPacketSize     0x0c00  2x 1024 bytes
    bInterval               1
Interface Descriptor:
  bLength                 9
  bDescriptorType         4
  bInterfaceNumber        1
  bAlternateSetting       6
  bNumEndpoints           1
  bInterfaceClass        14 Video
  bInterfaceSubClass      2 Video Streaming
  bInterfaceProtocol      0
  iInterface              0
  Endpoint Descriptor:
    bLength                 7
    bDescriptorType         5
    bEndpointAddress     0x81  EP 1 IN
    bmAttributes            5
      Transfer Type            Isochronous
      Synch Type               Asynchronous
```

```
        Usage Type              Data
      wMaxPacketSize     0x1380  3x 896 bytes
      bInterval               1
    Interface Descriptor:
      bLength                 9
      bDescriptorType         4
      bInterfaceNumber        1
      bAlternateSetting       7
      bNumEndpoints           1
      bInterfaceClass        14 Video
      bInterfaceSubClass      2 Video Streaming
      bInterfaceProtocol      0
      iInterface              0
      Endpoint Descriptor:
        bLength                 7
        bDescriptorType         5
        bEndpointAddress     0x81  EP 1 IN
        bmAttributes            5
          Transfer Type           Isochronous
          Synch Type              Asynchronous
          Usage Type              Data
        wMaxPacketSize     0x1400  3x 1024 bytes
        bInterval               1
  Device Qualifier (for other device speed):
    bLength                 10
    bDescriptorType         6
    bcdUSB               2.00
```

```
  bDeviceClass          239 Miscellaneous Device
  bDeviceSubClass         2 [unknown]
  bDeviceProtocol         1 Interface Association
  bMaxPacketSize0        64
  bNumConfigurations      1
Device Status:     0x0000
  (Bus Powered)
```

- **Query System & Kernel Attributes**

  (specific serial number or UUID for a udev rule or script)
  udevadm info --query=all --name=/dev/video2

```
P: /devices/pci0000:00/0000:00:14.0/usb1/1-9/1-9:1.0/video4linux/video2
M: video2
R: 2
U: video4linux
D: c 81:2
N: video2
L: 0
S: v4l/by-id/usb-Generic_ICT_Camera_200901010001-video-index0
S: v4l/by-path/pci-0000:00:14.0-usb-0:9:1.0-video-index0
S: v4l/by-path/pci-0000:00:14.0-usbv2-0:9:1.0-video-index0
E: DEVPATH=/devices/pci0000:00/0000:00:14.0/usb1/1-9/1-9:1.0/video4linux/video2
E: DEVNAME=/dev/video2
E: MAJOR=81
```

E: MINOR=2

E: SUBSYSTEM=video4linux

E: USEC_INITIALIZED=3082060137

E: ID_V4L_VERSION=2

E: ID_V4L_PRODUCT=ICT Camera: ICT Camera

E: ID_V4L_CAPABILITIES=:capture:

E: ID_BUS=usb

E: ID_MODEL=ICT_Camera

E: ID_MODEL_ENC=ICT\x20Camera

E: ID_MODEL_ID=5842

E: ID_SERIAL=Generic_ICT_Camera_200901010001

E: ID_SERIAL_SHORT=200901010001

E: ID_VENDOR=Generic

E: ID_VENDOR_ENC=Generic

E: ID_VENDOR_ID=0bda

E: ID_REVISION=0001

E: ID_TYPE=video

E: ID_USB_MODEL=ICT_Camera

E: ID_USB_MODEL_ENC=ICT\x20Camera

E: ID_USB_MODEL_ID=5842

E: ID_USB_SERIAL=Generic_ICT_Camera_200901010001

E: ID_USB_SERIAL_SHORT=200901010001

E: ID_USB_VENDOR=Generic

E: ID_USB_VENDOR_ENC=Generic

E: ID_USB_VENDOR_ID=0bda

E: ID_USB_REVISION=0001

E: ID_USB_TYPE=video

E: ID_USB_INTERFACES=:0e0100:0e0200:

E: ID_USB_INTERFACE_NUM=00

E: ID_USB_DRIVER=uvcvideo

E: ID_PATH_WITH_USB_REVISION=pci-0000:00:14.0-usbv2-0:9:1.0

E: ID_PATH=pci-0000:00:14.0-usb-0:9:1.0

E: ID_PATH_TAG=pci-0000_00_14_0-usb-0_9_1_0

E: ID_FOR_SEAT=video4linux-pci-0000_00_14_0-usb-0_9_1_0

E: COLORD_DEVICE=1

E: COLORD_KIND=camera

E: DEVLINKS=/dev/v4l/by-id/usb-Generic_ICT_Camera_200901010001-video-index0 /dev/v4l/by-path/pci-0000:00:14.0-usb-0:9:1.0-video-index0 /dev/v4l/by-path/pci-0000:00:14.0-usbv2-0:9:1.0->

E: TAGS=:snap_firefox_geckodriver:uaccess:snap_firefox_firefox:seat:

E: CURRENT_TAGS=:snap_firefox_geckodriver:uaccess:snap_firefox_firefox:seat: